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TECHNICAL ARTICLE

Friction Stir Powder Additive Manufacturing of Multi-track Depositions of Al 6061 Alloy for Cladding and 3D Printing Applications

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This study explores friction stir powder additive manufacturing of multi-track single-layer (MTSL) and multi-track multi-layer (MTML) depositions of aerospace grade Al 6061 for cladding and 3D printing applications respectively. Two MTSL depositions, each comprising of four deposition tracks with 2 mm and 4 mm overlap, were manufactured on Al 6061 substrate using the previously identified optimum process parameters. For 25 mm bending displacement, MTSL with 2-mm overlap failed at the interface between the adjacent tracks without showing any abrupt drop in bending force whereas 4 mm overlap MTSL did not fail because of stronger interface bonding. Therefore, 4 mm overlap was used to manufacture MTML deposition consisting of four depositions tracks and four deposition layers. MTML deposition failed at 17 mm bending displacement with an abrupt drop in bending force due to more deposition layers and their delamination. Bending force increased slightly because of regaining some strength from the substrate. Microhardness decreased marginally at overlapping regions in MTSL and MTML depositions compared with the non-overlapping regions because of localized high temperature caused by compressive forces. Tensile properties of Al 6061 alloy MTML along the track direction are comparable with corresponding single-track multi-layer deposition indicating good bonding between the tracks.

INTRODUCTION

Aluminum and its alloys are extensively used in aerospace, automobiles, and other engineering applications because of their high specific strength, stiffness, and low density.[1] Among different Al alloys, Al 6061 alloy is a widely used commercial alloy due to its high workability, weldability, and corrosion resistance.[2] Its additive manufacturing (AM) is being explored because of significant reduction in the material wastage, number of manufacturing steps, work-in-process, and number of distinct parts needed for an assembly.[3] Fusion-based AM (FBAM) processes such as selective laser melting[4] and wire arc additive manufacturing[5] melt the feedstock material for its deposition. However, it has detrimental effects on mechanical strength and structural performance of FBAM-made components from Al alloy due to various solidification problems such as porosity, hot cracking, and thermal distortion.[6] Solid-state AM (SSAM) processes have operating temperature in a range of 0.6-0.8 times the melting point of the feedstock material (i.e., no melting of feedstock material), thus eliminating problems related to the solidification. Recently, friction stir-based SSAM processes have attracted attention globally because of their inherent solid-state nature of deposition, high build rate, and

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wider material applicability. They are in high demand, particularly for Al alloys.[7] Different types of deposition are used for various applications: (1) single-track single-layer (STSL) deposition is commonly used for repairing, remanufacturing, and making minor adjustments to existing components,[8,9] (2) multi-track single-layer (MTSL) deposition with predefined overlaps between the adjacent track is used in different cladding applications,[10] (3) multi-track multi-layer (MTML) deposition is utilized for 3D printing applications and involves depositing multiple tracks and layers to achieve the desired width and height of deposition, respectively,[11] and (4) single-track multi-layer (STML) deposition is typically used for thin-walled geometries by depositing multiple layers on top of each other to attain the desired height.[12,13] It can also be used for repairing and remanufacturing applications and making some minor changes in the existing components.

Friction stir-based additive manufacturing (FSBAM) processes are still in their early stages of development with very few studies available on MTSL and MTML depositions by FSBAM processes. Dilip et al.[10] manufactured MTML depositions (consisting of 5 tracks and 3 layers) by multi-layer friction surfacing (MLFS) process using 19-mm-diameter mild steel rods and maintaining a 0.2 mm gap between the two consecutive tracks. They found that this process has larger deposition height and higher deposition rates than other direct metal AM processes. Gradual reduction in microhardness of the MTML deposition was observed from its top to the bottom deposition layer. Soujon et al.[14] manufactured MTML depositions (consisting of 2 tracks and 3 layers) of AA 5083-H112 (using it in a 20-mm-diameter rod) on AA 2050-T84 substrate by the MLFS process followed by the diffusion bonding. They used 2-mm overlap between two consecutive tracks. They found that diffusion bonding each time after the MLFS process reduces the defect to volume ratio up to 93%, and the most significant tunnel defect could be fully consolidated in the overlap area. Micro-flat tensile specimens extracted from different locations and orientations of the MTML did not reveal any significant directional gradient of tensile properties. Mukhopadhyay and Saha[15] manufactured MTML depositions (consisting of 4 tracks and 25 layers) of Al 6061 alloy powder on AA6061-T6 substrate by additive friction stir processing (AFSP) and found similar tensile properties along the directions of deposition tracks and deposition layers. Philips et al.[16] manufactured 228.6-mm-long and 69.9 mm-wide MTML depositions (comprising 2 tracks and 8 layers) of AA6061-T651 alloy (using it in 9.53 mm square cross-section rod) on AA6061-T651 substrate by the additive friction stir deposition (AFSD) process. They used 6.35-mm overlap between the consecutive tracks and a tool made of H13 hardened tool steel having four teardrop protrusions on its bottom face. They investigated location-specific microstructure, assessed tensile properties, and proposed a deposition equation for feedstock material feeding. They found that the bonding between two consecutive tracks is fully dense; there is no significant change in the microstructure of the overlapping region, and microhardness is reduced by 15 HV in the overlapping region.

Review of the relevant past work indicates that (1) no studies are available on MTSL depositions by FSBAM processes for cladding applications though some studies are available on MTML depositions by some FSBAM processes, (2) no studies are reported on MTSL and MTML using powder form of the feedstock material, and (3) overlap value plays a crucial role in multi-track depositions. Therefore, the present study aims to investigate the potential applications of friction stir powder additive manufacturing (FSPAM) process to manufacture MTSL depositions of Al 6061 alloy using different overlap values for cladding applications and MTML depositions for 3D printing applications. MTSL deposition was examined for its deposition geometry, bending strength, and microhardness whereas MTML deposition was studied for its deposition geometry, bending strength, microhardness, and tensile properties. Findings of the presented work will provide novel insights into the potential use of FSPAM process for solid-state cladding and 3D printing applications.

MATERIALS AND METHODS

Feedstock Material and FSPAM Apparatus

Powder of Al 6061 alloy, consisting of 1wt.% Mg, 0.8 wt.% Si, and balanced Al (obtained using energy dispersive spectroscopy) having powder particle size of 58 ± 20 μm was used as the feedstock material. Al 6061 alloy substrate plate of 200 mm length, 100 mm width, and 6 mm thickness was used for both MTSL and MTML type of depositions. Its chemical composition (as provided by the supplier) is listed in Table I. It was cleaned with acetone to remove any oxides prior to deposition.

Three-axis computer-controlled FSPAM apparatus was used to manufacture MTSL and MTML depositions of Al 6061 alloy. Its detailed description is available in Ref 17. It consists of a gravity-based powder feeder, which facilitates the supply of feedstock material ahead of a tilted rotating FSPAM tool. Argon gas was co-supplied along with the feedstock material powder at a rate of 0.2 normal liter (Nl) per minute to ensure its uniform flow. This value was selected because use of argon flow rate < 0.2 Nl/min was found to be inadequate for maintaining the uninterrupted supply of the feedstock material powder whereas use of a value > 0.2 Nl/min led to dispersion of the feedstock material powder away from the deposition zone. H13 tool steel was selected to manufacture the FSPAM tool of 22 mm diameter, and circumferential and radial

Table I. Chemical composition of Al 6061 alloy substrate plate

Constituent	Mg	Si	Cu	Cr	Mn	Zn	Fe	Al
Wt.%	0.8–1.2	0.4–0.8	0.15–0.4	0.04–0.35	Max 0.15	Max 0.25	Max 0.6	Balance

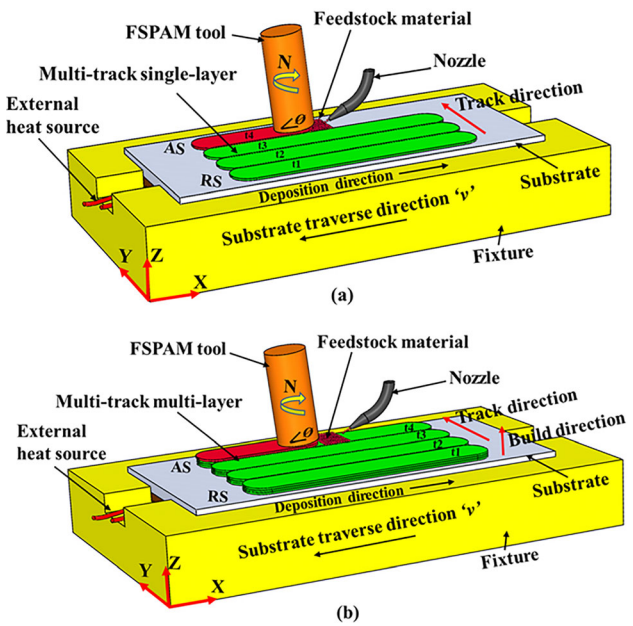

Fig. 1. Schematic of (a) multi-track single-layer deposition and (b) multi-track multi-layer deposition by the FSPAM process using external supply of Al 6061 powder (AS: advancing side, RS: retreating side).

grooves were imparted on its bottom face to ensure better consolidation of powder particles of feedstock material.

Procedure for Multi-track Single-Layer and Multi-track Multi-layer Depositions

Figure 1 schematically shows the procedure used in Al 6061 alloy MTSL (Fig. 1a) and MTML depositions (Fig. 1b) by FSPAM process using the previously identified optimum values[18] of feedstock material flow rate (as 460 mm^3/min), substrate traverse speed (as 25 mm/min), stand-off distance (SOD) between the tool and the substrate or an already deposited layer (as 0.5 mm), tool rotation speed (as 1200 rpm), and tool tilt angle (as 6°). A flat plate type external heat source preheated the substrate up to the artificial aging temperature of the feedstock material (which is 170°C for Al alloys) before beginning of deposition. Subsequently, the FSPAM tool mounted on a tilted spindle at 6 degrees is set to rotate at 1200 rpm. As the substrate is moved upward along the Z-direction, the interaction between the substrate and the FSPAM tool generates frictional heat, causing the interface temperature to exceed the recrystallization temperature of the feedstock material. Monitoring of the interface temperature is conducted using a thermal imaging camera (FLIR E95). The substrate is then moved downward by the SOD distance (0.5 mm). It is then traversed at a velocity of 25 mm/min along the *X*-direction while simultaneously feeding the feedstock material powder at a rate of 460 mm^3/min through an external nozzle positioned ahead of the FSPAM tool. This movement initiates stirring actions at the interfaces between the substrate, feedstock material, and FSPAM tool. These stirring actions induce severe plastic deformation within the feedstock material because of the compressive forces exerted on it. The feedstock material undergoes plasticization, leading to its deposition through metallurgical bonding with the substrate. Once a track of 130 mm length is deposited (depicted by 't1' in Fig. 1a), the substrate is moved downwards by SOD value (i.e., 0.5 mm). Feedstock material supply is temporarily halted until the substrate returns (along the *X*-axis) to the beginning point of a deposited track. The substrate is then moved along the *Y*-axis by a distance equivalent to the tool diameter minus the overlap value. This prepares the setup for the deposition of the next adjacent track ('t2' in Fig. 1a). This sequence is repeated for depositing the subsequent tracks 't3', 't4', and so on until the desired number of deposition tracks is manufactured to obtain multi-track single-layer (MTSL) deposition (as shown in Fig. 1a) for the cladding applications. The MTSL depositions obtained using 2 mm and 4 mm overlap are referred to as MTSL-2 and MTSL-4, respectively, in the subsequent text. Two depositions of each type were made to ensure reliability and consistency in the results.

Better overlap value obtained in MTSL deposition (i.e., 4 mm) was used to manufacture two distinct multi-track multi-layer (MTML) depositions consisting of four deposition tracks and four layers. It is referred to as MTML-4 in the subsequent text. The MTML deposition involves the integration of multiple MTSL depositions. Its deposition procedure is similar to that of MTSL deposition with the difference that the FSPAM tool returns to the initial position of the first deposition track each time for the beginning of subsequent MTSL deposition. Same procedure is used to deposit the desired number of layers along the build direction to get MTML-4 deposition as shown in Fig. 1b.

Preparation of Samples

Photographs of MTSL-2, MTSL-4, and MTML-4 types of depositions were taken prior to preparing

various samples from them to study their appearance. Four cuboid samples from MTSL-2, MTSL-4, and MTML-4 deposition each (i.e., total 12 samples) were cut by wire spark erosion machining (WSEM) process (Ecocut from Electronica, India). Dimensions of each cuboid sample were: 100 mm long covering all deposition tracks (along *Y*-direction), 10 mm wide (along *X*-direction), and total deposition height including the substrate (along *Z*-direction). Of these, cross section of one cuboid sample from each type of deposition was polished up to 2000 mesh size emery papers of silicon carbide. It was followed by fine polishing by 3 μm diamond paste and then 1 μm diamond paste to prepare the sample for studying deposition geometry parameters and microhardness. Substrate thickness for remaining three cuboid samples of MTSL-2, MTSL-4, and MTML-4 depositions was reduced by 3.5 mm from their bottom side, and their top surface was polished up to 800 mesh size silicon carbide emery papers to prepare samples of 2.9–3.0 mm height (or thickness) for MTSL-2 and MTSL-4 depositions, and 3.9–4.0 mm height for MTML-4 deposition for their 3-point bending test according to ASTM E290-14 standard. Three miniature samples (each sample has dimensions and shape as described by Zheng et al.[19]) were cut from the middle of MTML-4 deposition along its deposition tracks (i.e., along the Y-axis) by the WSEM process, as shown in Fig. 2a. Then, substrate was removed from these samples to get free standing thickness of 1.5 mm (i.e., or height along Z-axis). Subsequently they were polished up to 800 mesh size emery papers of silicon carbide to remove the surface irregularities from it. Figure 2b shows the left over MTML-4 deposition after cutting the tensile samples from it.

Measurement of Responses

Deposition Geometry Parameters

Macrostructure images of cuboid sample cut from each MTSL-2, MTSL-4, and MTML-4 depositions were taken by the stereo zoom microscope (EZ4 HD from Leica, Germany). The captured macrostructure images were fed to Image J software to measure their total deposition width along the track direction and total deposition height along the build direction. Magnified view of the interface between the two tracks and substrate within the overlapping region was observed through the scanning electron microscope (JSM-7610FPlus from JEOL, Japan) images.

Bending Strength

The three-point bending test was performed on the prepared samples of MTSL-2, MTSL-4, and MTML-4 depositions of Al 6061 alloy on the 25 kN bending test machine (Nano Plug and Play from BISS, India). Each sample was placed on two rollers mounted on two adjustable supports fixed to a movable fixture as shown in Fig. 3. Distance between the adjustable supports 'C' was kept at 54 mm. One roller mounted on the fixed fixture touches the center of the test sample. Movable fixture is provided an upward motion at 0.5 mm/min. Applied force 'F' (N) for bending criteria of 25 mm displacement was recorded to calculate bending stress 'σ' (N/mm^2 or MPa) using Eq. 1.

$$\sigma = \frac{3FC}{2wt^2} \tag{1}$$

where '*F*' is the applied force (N) for the specified bending criteria; '*C*' is the distance between the adjustable supports (mm); '*t*' and 'w' are thickness and width of the sample prepared for bending test (mm), respectively.

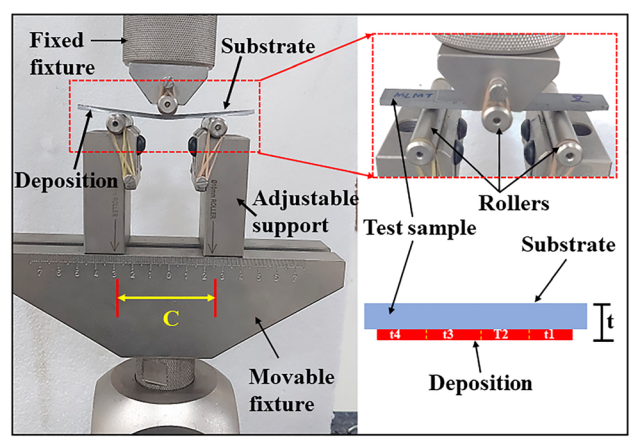

Fig. 3. Photograph of three-point bending machine along with a test sample.

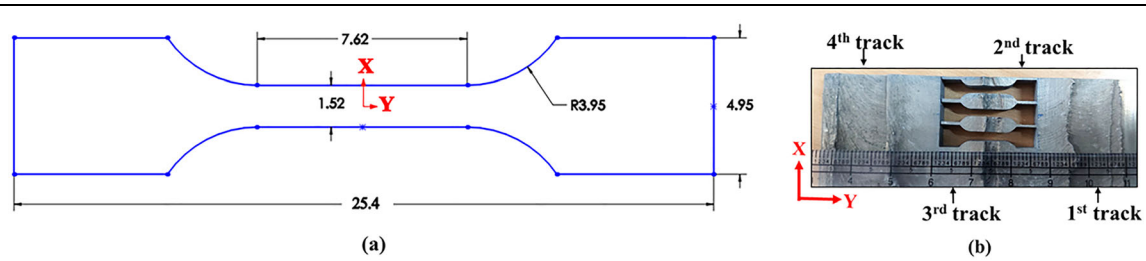

Fig. 2. Details of miniature tensile samples cut from Al 6061 alloy MTML-4 deposition for tensile test: (a) dimensions and (b) left over MTML-4 deposition after cutting tensile samples.

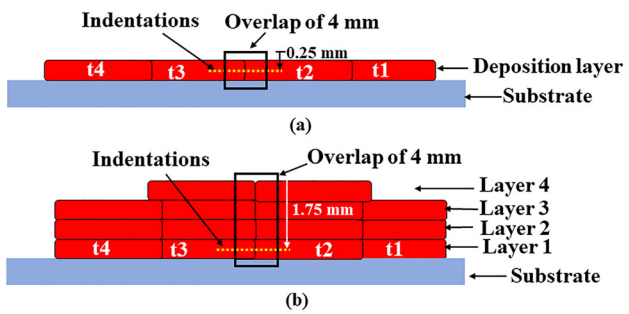

Fig. 4. Locations of the indentations made to measure of microhardness of sample of: (a) MTSL-4 deposition and (b) MTML-4 deposition.

The macrostructure images of the bent samples were taken by the stereo zoom microscope (EZ4 HD from Leica, Germany). These images were used to identify the cracks and fracture on the bending samples.

Microhardness

Microhardness of MTSL-4 and MTML-4 deposition samples was measured using the Vicker's microhardness tester (VMH-002 from Walter UHL, Germany) by applying 50 g load with 15 s dwell time following the ASTM-E384 standard. A total of 20 indentations were marked on 7.2-mm-long overlap region between 't2' and 't3' tracks at 0.25 mm below

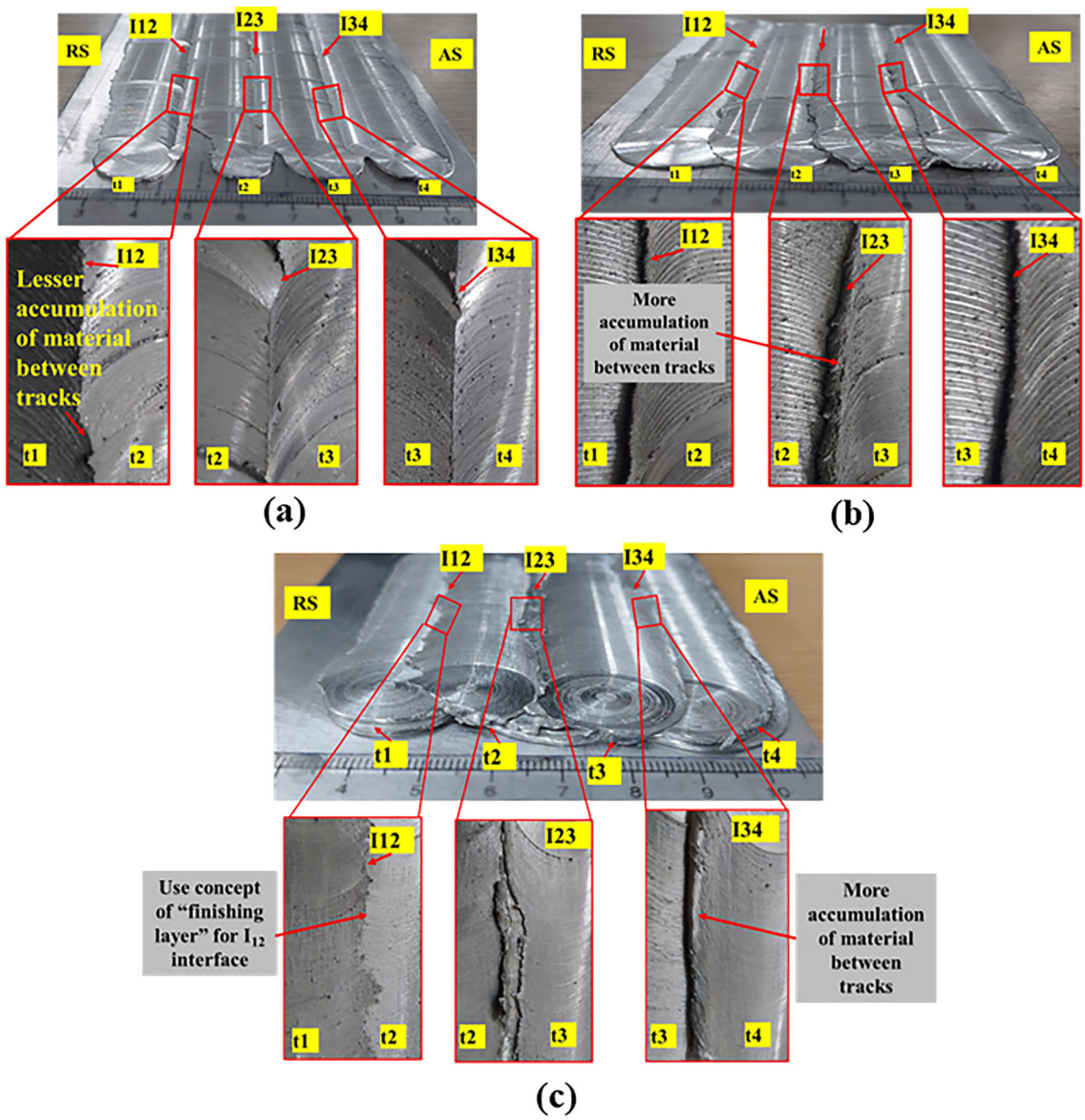

Fig. 5. Photographs showing top surface of FSPAM manufactured MTSL deposition of Al 6061 alloy for an overlap of (a) 2 mm and (b) 4 mm; (c) MTML deposition with 4 mm overlap (prefix 't' indicates deposition track and prefix 'I' indicates interface between two adjacent tracks).

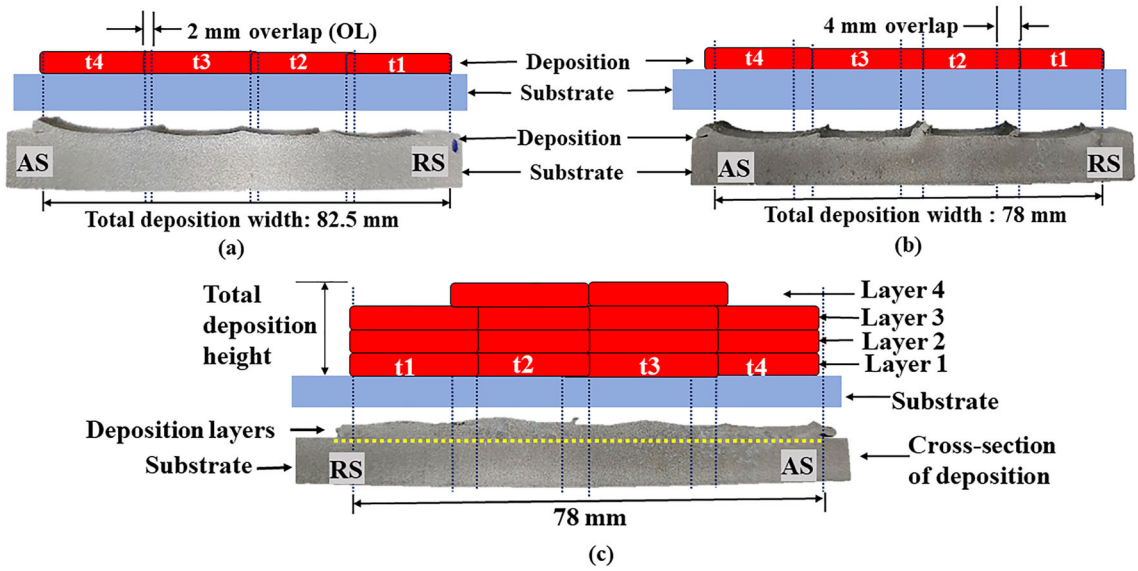

Fig. 6. Photographs showing cross section of FSPAM manufactured Al 6061 alloy multi-track depositions: (a) MTSL-2, (b) MTSL-4, and (c) MTML-4.

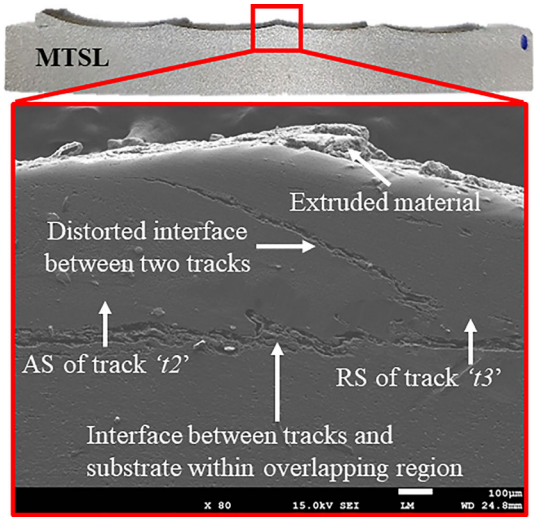

Fig. 7. SEM image showing magnified view of the interface between tracks and substrate within the overlapping region in a multi-track single-layer deposition.

the top surface of MTSL-4 deposition sample (as shown in Fig. 4a) and 1.75 mm below the top surface of MTML-4 deposition sample (as depicted in Fig. 4b).

Tensile Test for MTML-4 Deposition

Tensile tests of MTML-4 samples were performed on a 50 kN universal tensile testing machine (H50KL from Tinius Olsen, UK) at an elongation rate of 0.06 mm/min. The recorded data were used to plot stress-strain curve.

RESULTS AND DISCUSSION

Appearance and Deposition Geometry

Figure 5 depicts photographs showing appearance of top surface of MTSL deposition with 2 mm overlap, i.e., MTSL-2 (Fig. 5a), MTSL deposition with 4 mm overlap, i.e., MTSL-4 (Fig. 5b), and MTML deposition with 4 mm overlap, i.e., MTML-4 (Fig. 5c) along with their enlarged images of interfaces between two consecutive tracks. In these images, prefix 't' indicates deposition track and prefix 'I' indicates interface between two adjacent tracks, i.e., 'I34' indicates interface between deposition tracks 't3' and 't4' and so on. Figure 5 shows that MTSL-4 (Fig. 5b) and MTML-4 (Fig. 5c) depositions have significant material accumulation in the overlapping regions compared with the MTSL-2 deposition. This phenomenon can be attributed to extrusion of the material available at the overlapping region of the previous track when the subsequent adjacent track is deposited by the FSPAM tool.[16] Since SOD remains constant during the deposition of the adjacent track, the material at the overlapping region extrudes because of generation of compressive forces by the FSPAM tool. This material extrusion is more pronounced for higher overlap values and vice versa. The overlapping value affects microstructure and bending behavior of the claddings. It is important to note that the inherent nature of FSPAM process, which tends to accumulate more material on the advancing side (AS), is advantageous for the cladding applications. This is because the material accumulation on the AS of a deposition track is counterbalanced by the retreating side (RS) of the subsequent track resulting in a more uniform geometry of a cladding.

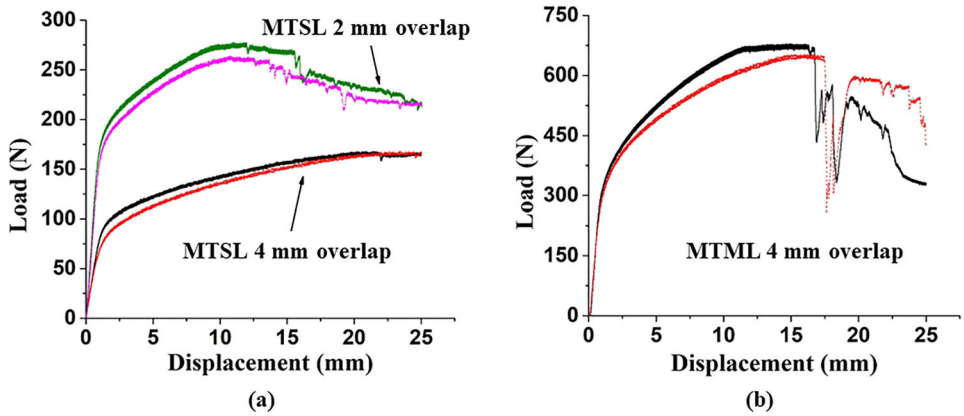

Fig. 8. Results of bending test depicting force-displacement curves for FSPAM manufactured Al 6061 alloy multi-track depositions: (a) MTSL-2 and MTSL-4 and (b) MTML-4.

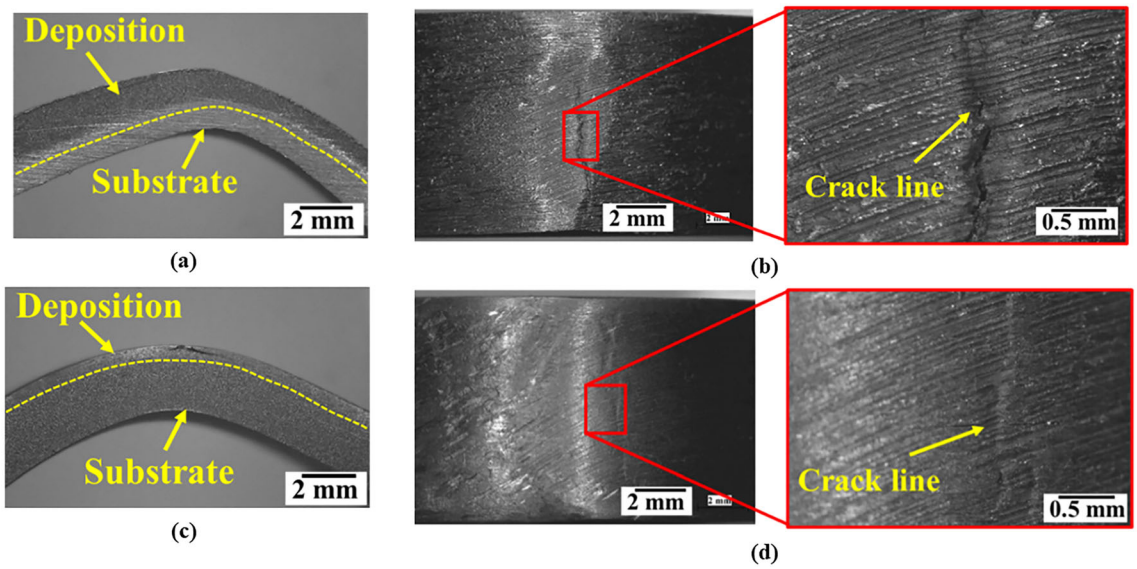

Fig. 9. Macrostructure images of bent samples of Al 6061 alloy MTSL-2 deposition: (a, c) side view and (b, d) top view.

Figure 6 depicts macrostructure showing cross-section of the MTSL-2 (Fig. 6a), MTSL-4 (Fig. 6b), and MTML-4 (Fig. 6c) deposition. These figures show that deposition of four tracks resulted in total deposition width as 82.5 ± 1.3 mm in MTSL-2, 78 ± 1.8 mm in MTSL-4, and 78 ± 2.4 mm in MTML-4 deposition. Total deposition height is 0.5 ± 0.05 mm and 2 ± 0.06 mm for STML and MTML, respectively. Lesser accumulation of material is clearly visible in cross section of MTSL-2 deposition as shown in Fig. 6a because of the smaller overlap value. The overlapping regions of MTSL-4 have slopy surfaces (as shown in Fig. 6b) due to larger overlap and extrusion of material available at the overlapping regions. Their removal needs post-deposition finishing or using the concept of a finishing layer between the two tracks by FSPAM process (i.e., using FSPAM tool without supplying the feedstock material). Finishing layer has significantly reduced accumulated material between the tracks as shown in Figs. 5c and 6c.

Figure 7 presents the SEM image showing a magnified view of the interface between tracks and substrate within the overlapping region in a typical multi-track single-layer deposition.

It is evident from the SEM image (Fig. 7) that the defects are not present at the interfaces between a deposition track and substrate and interface between the deposition tracks themselves. It indicates the strong bonding within the overlapping regions. Presence of depressions at the interface boundaries might have been caused by the chemical attack by the etchant.[20] The AS of the previously deposited track 't2' is restirred when subsequent track 't3' is deposited. The generation of the compressive forces due to overlapping has resulted in the distorted interface between the tracks as shown in Fig. 7. Moreover, it has partially extruded

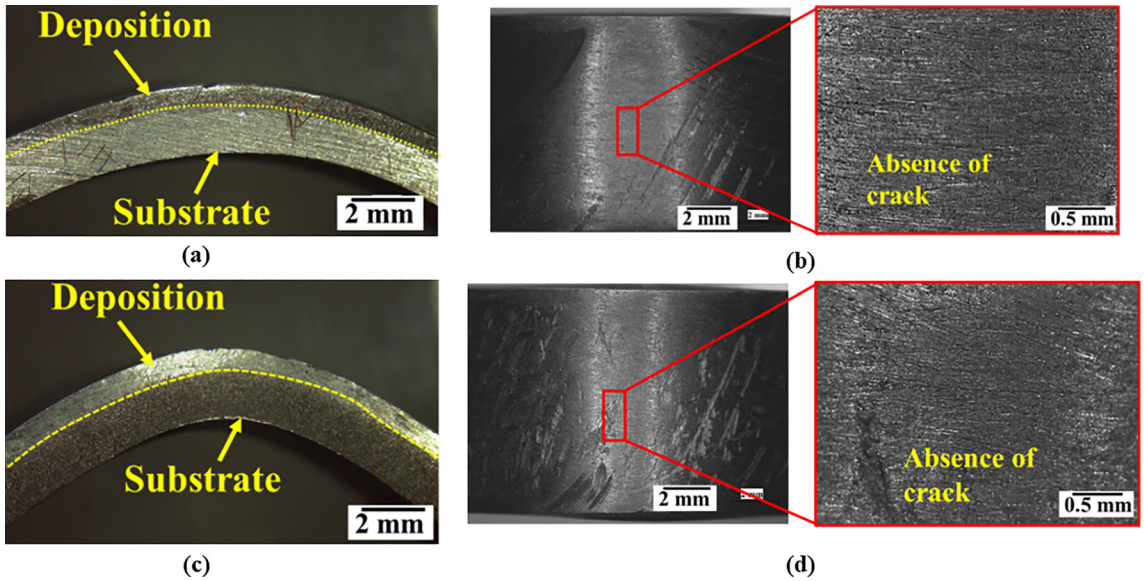

Fig. 10. Macrostructure images of bent samples for Al 6061 alloy MTSL-4 deposition: (a, c) side view and (b, d) top view.

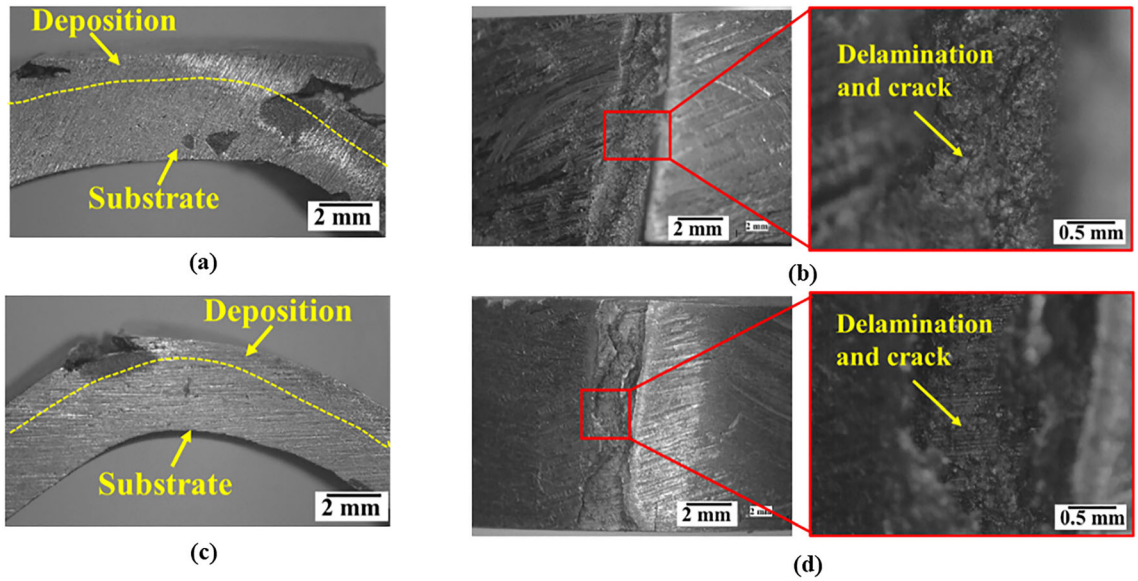

Fig. 11. Macrostructure images of bent samples for Al 6061 alloy MTML-4 deposition: (a, c) side view and (b, d) top view.

material of the track 't2' and is replaced by new material of the track 't3'. The extruded material has accumulated on top of the deposition tracks within the overlapping region as shown in Fig. 7 which needs to be removed for better finishing. The localized heat in the FSPAM process does not alter the bulk properties of the substrate[16] thus avoiding dilution or major microstructural changes within the overlapping regions. However, the restirring effect under severe deformation of powder particles may lead to dissolution of Mg_2Si precipitates within the overlapping regions causing variation in its mechanical properties compared to the non-overlapping regions.

Results of Three-Point Bending Test

Figure 8 shows results of three-point bending test depicting bending force-displacement curves for MTSL-2 and MTSL-4 (Fig. 8a) and MTML-4 (Fig. 8-b) depositions. Figures 9, 10, and 11 show macrostructures of two bent samples each for MTSL-2 (Fig. 9a–d), MTSL-4 (Fig. 10a–d), and MTML-4 (Fig. 11a–d) depositions showing their side and top view.

Figure 8a shows that bending force increases sharply with displacement in MTSL-2 sample during initial stage, which corresponds to the elastic deformation of the bending sample. However, further increase in displacement increases the bending

force at a reduced rate, and it attains peak value of 271 N (at a displacement of approximately 11.5 mm) because of work hardening during the plastic deformation.[21] Bending stress of MTSL-2 deposition was found to be 252 MPa. Thereafter, the bending force gradually decreases in MTSL-2 deposition because of the propagation of crack as shown in Fig. 9b and d. However, bending force did not decrease in case of MTSL-4 deposition (as shown in Fig. 8a) for the displacement criterion of 25 mm implying that (i) it has higher bending strength than MTSL-2 deposition, which is due to absence of cracks, as shown in Fig. 10b and d, i.e., it requires higher displacement for its bending failure; (ii) overlap of four mm helps in stronger metallurgical bonding between the tracks. Bending force decreases suddenly for MTML-4 deposition after attaining its maximum value (i.e., 668 N) at approximately 17 mm displacement (as shown in Fig. 8b), which is due to delamination of layers depicted in Fig. 11b and d. The bending stress of MTML-4 was found to be 346 MPa at the maximum bending force. During the initial stages of bending, different layers in MTML-4 deposition provide more resistance to the deformation thereby increasing bending force at a faster rate. Once the bending force reaches a certain magnitude, crack formation is observed at the interfaces between the tracks where it experiences tensile stress. The crack expands and propagates till it reaches the substrate side as the displacement increases. The bending force increases again, which might be due to regaining of some strength from the unbroken substrate.[22] Fan et al.[23] have mentioned that the inter-granular fracture can cause fracture in bending test of Al-Mg-Si alloy sheets. Such inter-granular fracture depends on presence of precipitates at the grain boundaries and associated precipitate-free zones.

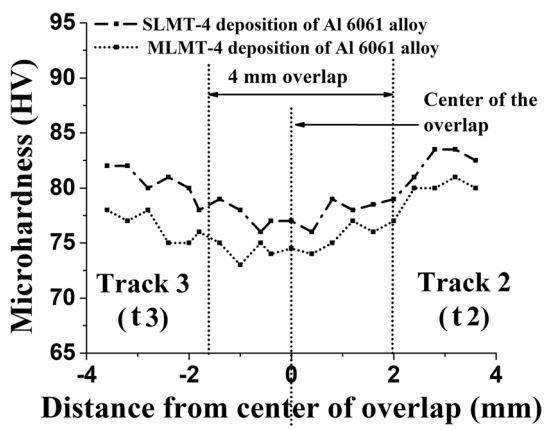

Fig. 12. Microhardness values along the cross-section of deposition tracks in the first deposition layer of Al 6061 alloy MTSL-4 and MTML-4 depositions.

Mechanical Properties

Figure 12 depicts microhardness values along the cross-section of deposition tracks in the first deposition layer of Al 6061 alloy MTSL-4 and MTML-4 depositions. Their average values (taken over overlapping regions in 4 mm) of microhardness are 77.9 HV and 75.1 HV, respectively. Comparatively smaller value of microhardness in the first deposition layer of MTML-4 deposition than MTSL-4 deposition is due to experience of more thermal cycles, which causes dissolution of precipitates as observed in FSPAM manufactured Al 6061 alloy STML depositions.[17] Similar observations have been reported by Philips et al.[24] for Al-Mg-Si alloy multi-layer deposition by the AFSD process. Microhardness decreases marginally at the overlapping regions compared with the non-overlapping regions in both MTSL-4 and MTML-4 depositions as depicted in Fig. 12. Similar observations have been made by Philips et al.[16] for Al-Mg-Si alloy two-track eight-layer deposition by the AFSD process. While

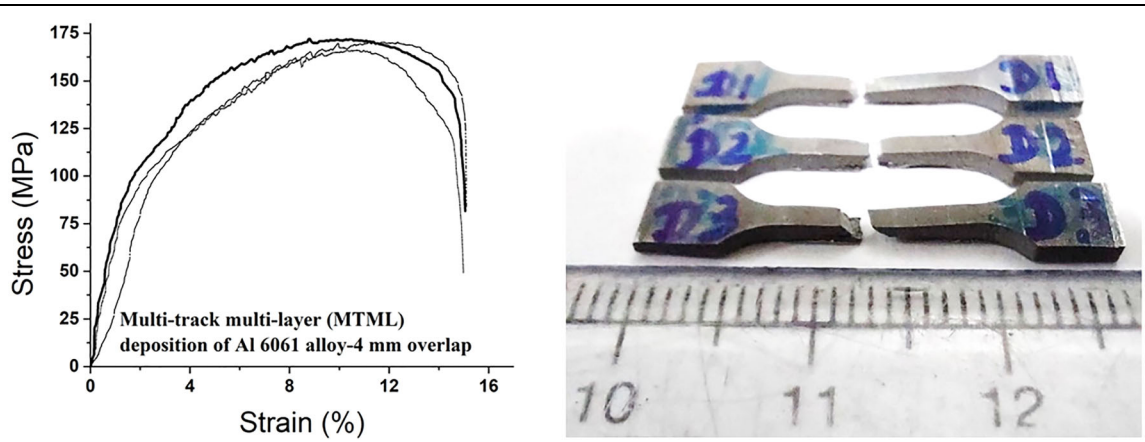

Fig. 13. Tensile properties of FSPAM manufactured Al 6061 alloy MTML-4 deposition: (a) stress-strain graph and (b) fractured samples in the tensile test.

Table II. Tensile properties of Al 6061 alloy MTML-4 deposition

Deposition	Yield strength (MPa)	Ultimate tensile strength (MPa)	Elongation (%)
MTML-4	95.8 ± 2.2	169.4 ± 3	15 ± 0.06

depositing subsequent tracks, the feedstock material at the overlapping regions undergoes extensive plastic deformation caused by high compressive forces. It creates high friction, which increases temperature at the overlapping regions which dissolves the precipitates and decreases the microhardness.

Figure 13 presents tensile properties of FSPAM manufactured MTML-4 deposition showing stress-strain graphs belonging to three replicates of the same sample (Fig. 13a) and their fractured images (Fig. 13b). Table II presents its values of % elongation and yield and ultimate tensile strengths. Figure 13a and Table II show that yield, ultimate tensile strength, and % elongation of MTML-4 deposition are 95.8 ± 2.2 MPa, 169.4 ± 3 MPa, and 15 ± 0.06%, respectively. These values are comparable to the tensile properties of FSPAM manufactured STML of Al 6061 alloy measured along its deposition direction.[17] This indicates strong bonding at the overlapping regions between the tracks in MTML-4 deposition.

CONCLUSION

This study explored multi-track single-layer and multi-layer multi-track depositions of Al 6061 alloy by FSPAM process for potential applications in cladding and 3D printing. More attention was directed towards importance of the overlap values and corresponding influence on the bending strength of the depositions. The following are the conclusions from this work:

- The MTSL-4 deposition showed slopy surfaces due to larger overlap value and extrusion of material available at the overlapping regions. It requires post-deposition finishing.
- Inherent nature of FSPAM process to accumulate more material on AS is beneficial in cladding applications as material accumulation on AS of a deposition track is balanced by the RS of the next track.
- Application of finishing layer between the two tracks by FSPAM process, i.e., using FSPAM tool without supplying the feedstock material, was found to significantly reduce the material accumulated between the tracks.
- The MTSL-4 depositions sustained higher bending force than MTSL-2 depositions because of absence of cracks and stronger metallurgical bonding between the tracks.
- The MTSL-2 deposition showed bending stress of 252 MPa and was found to be fractured at a displacement of 11.5 mm because of crack propagation whereas MTSL-4 sustained a set displacement of 25 mm without fracture. The MTML-4 deposition resulted in maximum bending strength of 346 MPa at approximately 17 mm displacement.
- Microhardness decreased marginally at the overlapping regions because of higher temperature caused by localized severe deformation in both MTSL and MTML depositions.
- The MTML-4 deposition showed comparable tensile properties to corresponding STML depositions indicating good bonding at the overlapping regions between the deposition tracks.
- Findings of this work prove that FSPAM process is capable of imparting good metallurgical bonding between deposition tracks in MTSL and MTML of aerospace grade Al 6061 alloy, which are useful in their cladding and 3D printing applications.

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AUTHOR CONTRIBUTIONS

All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Bhavesh Chaudhary. The first draft of the manuscript was written by Bhavesh Chaudhary, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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DATA AVAILABILITY

Authors understand that our manuscript and associated personal data will be shared with Research Square for the delivery of the author dashboard.

CODE AVAILABILITY

No software has been developed in this manuscript.

CONFLICT OF INTEREST

The authors have no conflicts of interest to declare that are relevant to the content of this article.

CONSENT FOR PUBLICATIONS

All the authors have given their consent for publication of this manuscript.

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